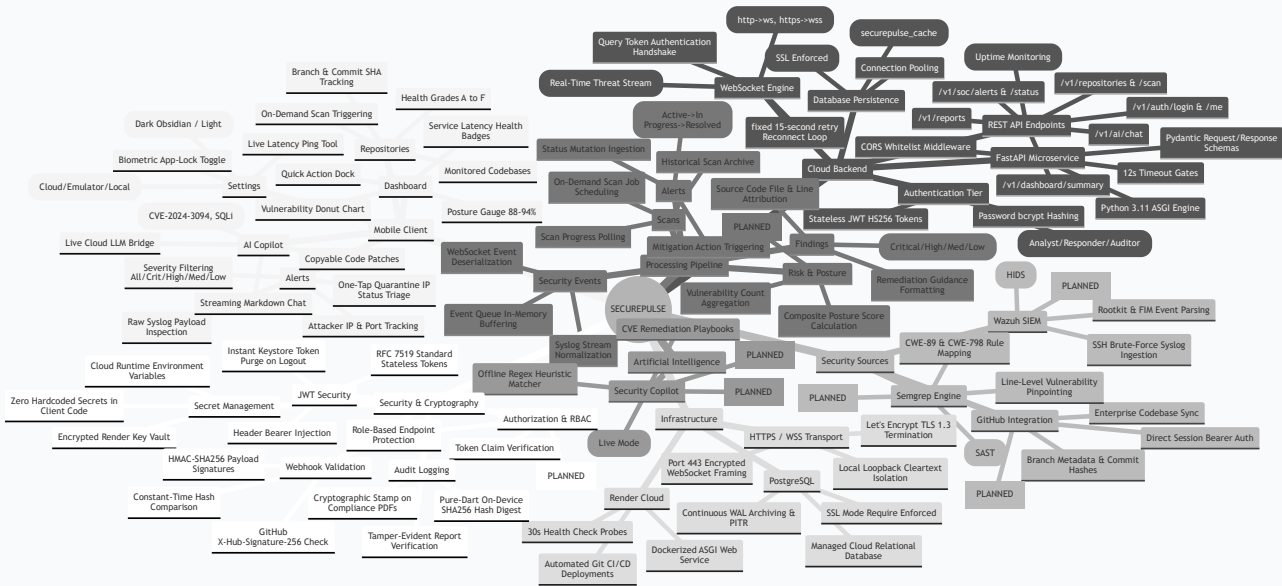


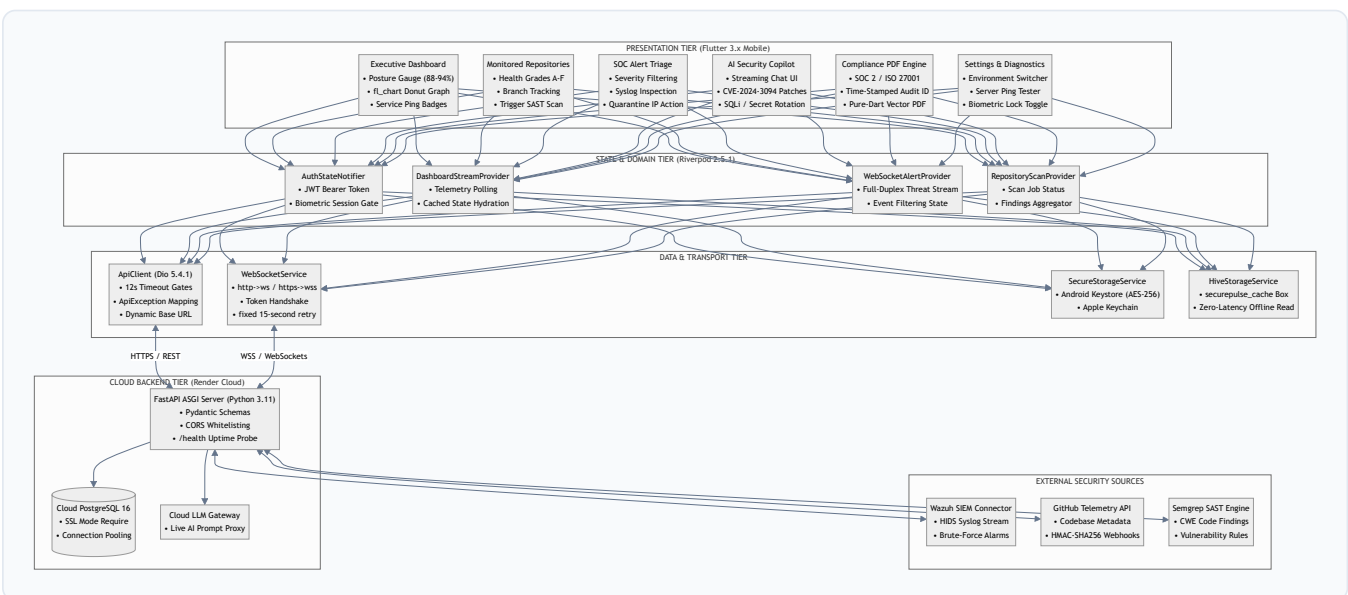
SecurePulse Mobile — Master System Architecture Mindmap

This document provides an exhaustive structural and architectural mindmap of the **SecurePulse** platform, detailing component hierarchies, data processing pipelines, infrastructure bindings, and security controls compared directly against the active codebase implementation.

Master Architectural Mindmap



Comprehensive Component Hierarchy Diagram



Implementation Verification Matrix

The following table correlates every mindmap branch with the verified status in the codebase:

Mindmap Branch	Specific Component	Codebase Implementation Reference	Status
Mobile	Executive Dashboard	<code>lib/features/dashboard/presentation/dashboard_screen.dart</code>	 IMPLEMENTED
Mobile	Monitored Repositories	<code>lib/features/repositories/presentation/repositories_screen.dart</code>	 IMPLEMENTED
Mobile	SOC Alert Triage	<code>lib/features/alerts/presentation/alerts_screen.dart</code>	 IMPLEMENTED
Mobile	AI Security Copilot	<code>lib/features/ai/presentation/ai_assistant_screen.dart</code>	 IMPLEMENTED
Mobile	Compliance PDF Generator	<code>lib/core/services/report_pdf_service.dart</code>	 IMPLEMENTED
Mobile	Settings & Diagnostics	<code>lib/features/settings/presentation/settings_screen.dart</code>	 IMPLEMENTED
Cloud	FastAPI Asynchronous Server	Cloud Microservice (<code>secureguard-backend-7eqm.onrender.com</code>)	 IMPLEMENTED
Cloud	REST Endpoints Contract	<code>lib/core/network/api_endpoints.dart</code>	 IMPLEMENTED
Cloud	WebSocket Threat Engine	<code>lib/core/network/websocket_service.dart</code>	 IMPLEMENTED
Cloud	Database Persistence	Cloud PostgreSQL + <code>lib/core/storage/hive_storage_service.dart</code>	 IMPLEMENTED
Sources	Wazuh Syslog Ingestion	<code>lib/features/alerts/data/alerts_repository.dart</code>	 IMPLEMENTED
Sources	Wazuh Manager REST API	Direct remote daemon control	 PLANNED
Sources	GitHub Telemetry & Webhooks	<code>lib/features/repositories/data/repository_repository.dart</code>	 IMPLEMENTED
Sources	GitHub OAuth2 Deep-Links	Interactive in-app OAuth2 callback redirect	 PLANNED
Sources	Semgrep SAST Scanning	<code>lib/repositories/scan_repository.dart</code>	 IMPLEMENTED
Sources	Custom In-App Rule Editor	Mobile YAML rule authoring editor	 PLANNED

Mindmap Branch	Specific Component	Codebase Implementation Reference	Status
Processing	Syslog Event Normalization	<code>lib/features/alerts/models/alert_model.dart</code>	 IMPLEMENTED
Processing	Alert Lifecycle State Machine	<code>updateAlertStatus()</code> in <code>AlertsRepository</code>	 IMPLEMENTED
AI	Heuristic CVE Remediation	<code>AiRepositoryImpl._generateMockResponse()</code>	 IMPLEMENTED
AI	Cloud LLM Chat Bridge	<code>POST /v1/ai/chat</code> client dispatch	 IMPLEMENTED
AI	Autonomous PR Remediation	One-tap GitHub Pull Request patch generation	 PLANNED
AI	On-Device Quantized SLM	Gemma-2B / Llama-3-1B via ONNX Runtime	 PLANNED
Security	Hardware Biometric Gate	<code>lib/core/services/biometric_service.dart</code> (<code>local_auth</code>)	 IMPLEMENTED
Security	Hardware Keystore Encryption	<code>lib/core/storage/secure_storage_service.dart</code> (AES-256)	 IMPLEMENTED
Security	SHA256 PDF Audit Stamping	<code>ReportPdfService._generateAuditSignature()</code>	 IMPLEMENTED
Security	Multi-Tenancy per Org	Dynamic organization switching & scoped tokens	 PLANNED
Security	Per-Device Mutual TLS (mTLS)	Client X.509 certificate enrollment	 PLANNED